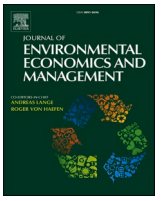
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Air pollution and the airborne diseases: Evidence from China and Japan☆

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ABSTRACT

This paper estimates the impact of ambient air pollution on the two most economically costly airborne respiratory diseases, COVID-19 and influenza. Our methods incorporate the epidemiological Susceptible-Infectious-Recovered-Deceased (SIRD) model to construct the outcome of interest, the Instrumental Variable (IV) model to establish causality, and the Flexible Distributed Lag (FDL) model to capture dynamic effects. Analyzing data from the first wave of the COVID-19 pandemic in China, we find that air pollution can significantly raise the daily growth rate of COVID-19. In contrast, air pollution shows small and statistically insignificant effects on influenza healthcare visits in Japan.

1. Introduction

The majority of the world's population breathes air that exceeds WHO guideline limits, leading to a grave threat to human well-being (World Bank, 2022). Decades of research have documented that air pollution increases morbidity and mortality, particularly among children and older adults (Graff Zivin and Neidell, 2013). In addition to physical health, it is also well documented that air pollution adversely impacts labor productivity, cognitive ability, mental health, emotion, and even the suicide rate (Aguilar-Gomez et al., 2022; Molitor et al., 2023; Yamada and Narita, 2020; Zheng et al., 2019). In response, individuals spend a substantial portion of their incomes on preventive investments that would not otherwise occur, such as medications (Deschênes et al., 2017), masks (Zhang and Mu, 2018), and air filters (Ito and Zhang, 2020). As textbook optimal policy requires setting the corrective tax equal to the social marginal damages, investigating an extensive set of outcomes that are relevant to human well-being and are affected by air pollution is

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of great policy relevance.

Scientists have been speculating that ambient air pollution could affect the spread of airborne disease, a global public health burden that, at times, can kill millions of populations. First, a rich line of literature has documented that air pollution can cause a persistent inflammatory response and impair the respiratory and immune systems (Ciencewicki and Jaspers, 2007; Diamond et al., 2000), making it more challenging for an individual to resist infections. Second, studies suggest that aerosols in the air may maintain the viability and transmissibility of the virus (van Doremalen et al., 2020; Liu et al., 2020). Therefore, degraded air quality, dominated by particulate matter or aerosols, may extend the survival of the virus in the air, which increases individuals' exposure to the virus. Finally, air pollution may prompt individual behavioral changes that alter susceptibility to disease transmission. People might adjust the amount of time spent indoors and outdoors, affecting their proximity to those already infected and, consequently, the likelihood of infection. Not surprisingly, a thin but growing literature has documented that air pollution leads to increased infections of SARS (Xiang et al., 2014), influenza (Clay et al., 2018; Graff Zivin et al., 2023), tuberculosis (Escombe et al., 2007), chickenpox (Leclair et al., 1980), and SARS-CoV-2 (i.e., COVID-19) (Austin et al., 2023; Isphording and Nico, 2021; Persico and Johnson, 2021), while a few recent studies report the null impacts (Conte et al., 2024; Provenzano et al., 2024).

Estimating the relationship between ambient air pollution and airborne diseases is challenging for three reasons. First, the epidemiological characteristics of viruses can differ significantly from other health outcomes. Viruses grow exponentially, leading to widespread infections within a short time, a pattern not typically observed in mortality and morbidity data. Second, it is well documented in the economics literature that air pollution levels are endogenous to various health outcomes (Arceo et al., 2016; Chay and Greenstone, 2003; Cheung et al., 2020; Deryugina et al., 2019; Deschênes et al., 2017; Fan et al., 2020; Heft-Neal et al., 2020; Knittel et al., 2016). Lastly, the contemporaneous relationship between air pollution and new infections can be distorted due to the delays caused by incubation, testing, and reporting.

To overcome these challenges and estimate the relationship between ambient air pollution and airborne diseases, we integrate a well-established epidemiological model, a causal inference framework, and non-linear dynamic specifications, namely the Susceptible-Infectious-Recovered-Deceased (SIRD) model, the Instrumental Variable (IV) model, and Flexible Distributed Lag (FDL) model, respectively.

In our specifications, we use the growth rate of confirmed active infections, rather than total confirmed cases, as the outcome variable. This is because, in the SIRD model, the growth rate of active infections is proportional to the transmission rate, a key parameter that dictates virus spread and its non-linear dynamics. Second, to address the endogeneity issues associated with air pollution, we leverage plausibly random variation in thermal inversion as an instrumental variable (e.g., Arceo et al., 2016; He et al., 2019). When a thermal inversion occurs, a warmer air layer overlays a cooler layer near the ground, trapping pollutants emitted from the surface and leading to elevated local air pollution. As this phenomenon results from complex meteorological processes, it provides quasi-random variation in ambient air pollution levels. Finally, to account for potential lags between infections and case confirmations, we incorporate the FDL model to approximate these delayed effects using finite-order B-spline functions (Barwick et al., 2023).

We apply the econometric model to two case studies: COVID-19 in China and influenza in Japan, as both countries provide comprehensive public datasets on disease, air pollution, and weather information. Importantly, most previous studies on this topic use data from the U.S. and selected countries in Europe and few of them focus on Asia (see Table 1). Given the significant cultural and socioeconomic differences across different countries, it is important to examine the relationship between air pollution and airborne diseases in broader research settings. Economically, our analysis provides insights into the effects of air pollution on infectious diseases in the two largest Asian economies, which together account for approximately 20% of global GDP.

In the first case study, we use COVID-19 infection data from the initial outbreak in China, where governments made enormous efforts to collect timely data to control the spread of the disease. We find that air pollution indeed facilitates the transmission of COVID-19: a 10-point (14.3%) increase in the Air Quality Index (AQI) would lead to a 2.93 percentage point increase in the daily COVID-19 growth rate, with a delay of 2–13 days, consistent with epidemiological observations regarding incubation and symptom periods (corresponding to a 0.15–0.17 increase in the reproduction number, R_0). A back-of-the-envelope calculation suggests that if China had managed to reduce the daily AQI to below 100 in every city on each day during our study period (thus reducing the mean AQI by 13.2%), the total confirmed cases would have decreased by 27.0% (from 30,376 to 22,181). In this simulation, air pollution reshaped the exponential growth of the virus: the difference in the confirmed cases between the two scenarios (observed vs. below 100) is initially small but gradually becomes larger as the outbreak progresses, a pattern that cannot be observed without incorporating epidemiological characteristics.

In the second case study, we assess how ambient air pollution alters the transmission of influenza using data from Japan from 2010 to 2018 by analyzing prefecture-by-week influenza data from approximately 5500 representative hospitals. Unlike our findings for COVID-19, we find small and statistically insignificant effects of pollution on influenza growth rates using both the OLS and IV estimators. Although the point estimate still has the expected sign—specifically, a 10% increase in SPM (1.84 $\mu g/m^3$) leads to a 2.5 percentage point increase in the growth rate in the same week when using the IV estimator—this estimate is statistically indistinguishable from zero. Including lags, adding more flexible fixed effects, or restricting samples to the flu seasons yield similar insignificant results.

Existing economic research presents mixed evidence of the effects of air pollution on various respiratory infectious diseases. In the case of the COVID-19 pandemic, most studies find that higher air pollution concentrations increase infections,[1] with a few exceptions

[1] See Table 1 for the summary of existing findings on how air pollution affects airborne diseases. Besides, several papers estimate the effects of long-term exposure to air pollution on COVID-19 cases and mortality (Cole et al., 2020; Tanaka, 2024). Additionally, medical research investigates the relationship between climatic conditions, air pollution, and COVID-19 outcomes (e.g., Xu et al., 2021).

Table 1
Summary of the existing papers on air pollution and infectious diseases.

Authors	Country	Data	Air Pollution	IV	Findings
Panel A: COVID-19					
This Paper	China	city-by-day	AQI	thermal inversion	A 10-point increase in AQI increases the daily growth rate of COVID-19 by 2.93 percentage points
Provenzano et al. (2024)	US	county/state-by-week	AQI	thermal inversion	No relationship between air quality and either COVID-19 or ILI cases.
Conte et al. (2024)	US	census tract level	PM2.5, NO2, NO	wind direction	No causal impacts of increased chronic concentrations of traffic-related air pollutants on COVID-19 death or hospitalization counts.
Austin et al. (2023)	US	county-by-day	PM2.5	wind direction	A 1 $\mu g/m^3$ increase in $PM_{2.5}$, or 15% of the average $PM_{2.5}$ concentration in a county, increases the number of same-day confirmed cases by 1.8% and the same-day death rate by just over 4% from the mean in a county.
Isphording and Pestel (2021)	Germany	county-by-day	PM10, PM2.5, Ozone	wind direction	PM_{10} concentration significantly positively affects death numbers from four days before to ten days after the onset of symptoms.
Persico and Johnson (2021)	US	county-by-day	PM2.5, Ozone	NA	Following the EPA's rollback of enforcement, counties with more Toxic Release Inventory sites saw an average 11.8 percent increase in pollution. These policy-induced increases in pollution are associated with a 53 percent increase in cases and a 10.6 percent increase in deaths from COVID-19.
Panel B. Influenza					
This Paper	Japan	prefecture-by-week	SPM	thermal inversion	The impact of Suspended Particulate Matter on influenza growth rate is small and statistically insignificant.
Graff Zivin et al. (2023)	US	county-by-month	AQI	wind direction	Increased levels of rise in contemporaneous pollution influenza hospitalizations.
Provenzano et al. (2024)	US	county/state-by-week	AQI	thermal inversion	No relationship between air quality and either COVID-19 or Influenza Illness cases.

suggesting no impacts (Conte et al., 2024; Provenzano et al., 2024). Within those positive-impact studies, the magnitude of the pollution impacts varies significantly across different studies. For example, Austin et al. (2023) show that a 1-$\mu g/m^3$ increase in PM2.5 (15% of mean) raises the same-day confirmed cases by 1.8% in the U.S. Isphording and Pestel (2021) observe that a 1-$\mu g/m^3$ increase in PM10 (6.9% of mean) results in a 13–18% increase in confirmed cases in Germany, somewhat comparable to our findings from China. In contrast, Persico and Johnson (2021) report a much larger impact: they show that US EPA's environmental deregulation during COVID is associated with a 0.778-$\mu g/m^3$ increase in PM2.5 (11.8%) and a 53.0% increase in daily confirmed cases. For influenza, Graff Zivin et al. (2023) show that contemporaneous pollution increases influenza hospitalizations, particularly among unvaccinated populations. However,Provenzano et al. (2024) find a null result.

Our paper differs from previous studies in many dimensions. In terms of data, for example, the disease variability, infectious stages, vaccination rates, herd immunity, and pharmaceutical interventions in China and Japan are all different from those in the U.S. and Europe, not to mention that population characteristics, levels of economic development, and institutions also vary considerably. While these differences make it difficult for us to draw meaningful conclusions on what specific factors could explain the mixed results, they also provide a valuable addition to the literature by documenting how air pollution affects airborne diseases in two large countries which together have more than one-fifth of the world population.

Methodologically, the key difference between our study and previous ones is that we estimate the key parameters by combining the SIRD model with causal inference models. The advantage of incorporating the SIRD model in the estimation is that the model can better capture the non-linear progression of viral spread and identify the specific channels through which air pollution influences disease transmission. In fact, several studies have combined the SIRD model with a causal inference framework and evaluated the effects of non-pharmaceutical interventions (Chernozhukov et al., 2021; Hsiang et al., 2020), masking (Karaivanov et al., 2021), and climate conditions (Carleton et al., 2021) on COVID-19. We extend this approach to estimate the impacts of air pollution on the transmission of airborne diseases.

Nevertheless, our approach relies on several assumptions that tend to oversimplify reality. In particular, it assumes that the entire population is susceptible to infection and that air pollution only affects virus transmission, not recovery and mortality rates, both of which can be violated in reality. It is also noted that the SIRD model, more generally, assumes a homogeneous population and uniform mixing and does not account for demographic variations, spatial heterogeneity, or changes in individual behavior (e.g., Acemoglu et al., 2021). Given these controversies, we see our findings as a complement to previous studies rather than a substitute that can give a firm answer to this important topic.

Broadly speaking, our work also adds to the growing body of literature on the impacts of air pollution on various health outcomes in China and Japan. Previous studies have investigated the effects of air pollution on life expectancy (Ebenstein et al., 2017), mortality (Fan et al., 2020; He et al., 2016; Kang et al., 2024), and morbidity (Barwick et al., 2023). More broadly, we contribute to the literature on how various environmental risks affect health outcomes in China and Japan (Akesaka and Shigeoka, 2023; Fu et al., 2023; He and Tanaka, 2023; Neidell et al., 2021).

2. Scientific background

Ambient air pollution can affect the spread of infections by increasing individuals' susceptibility and exposure to the virus. First, short-term exposure to severe air pollution can weaken the immune system's response to the virus and thereby increase the chance of infection (Ciencewicki and Jaspers, 2007; Diamond et al., 2000). In physiological studies, inhaled particulates and other air pollutants can interact with immune cells within the airways. It is found that pollutants can trigger cellular signaling pathways, leading to multicellular immune responses and perturbation, and eventually cause disease or fail to prevent disease (Ghio et al., 2012). In epidemiological studies, air pollutants such as PMs, NO_2, SO_2, and O_3 are also found to be associated with inflammatory and immune responses (Tellier et al., 2019; Vawda et al., 2014).

Second, aerosols in air pollutants may affect virus activity and facilitate its transmission. Aerosols are suspensions of solid or liquid particles in the air. These particles are small and have a low settling velocity; therefore, they can remain airborne for prolonged periods. For example, coughing and sneezing can generate a substantial quantity of particles. Because of evaporation, a large number of these particles shrink and then behave as aerosols. Aerosols due to coughing, sneezing, and breathing can result in virus transmission, in which the virus from an infected person can be carried over a fairly long distance, compared to the transmission through large droplets and direct contact.

Additionally, air pollution may induce behavioral changes that influence susceptibility to disease transmission. Research shows that air pollution affects outdoor activities (Moretti, 2011; Barwick et al., 2024b), short-term migration (Chen et al., 2022; Barwick et al., 2024a), and defensive behaviors such as mask-wearing (Zhang and Mu, 2018) and the use of air filters (Ito and Zhang, 2020). Among these behavioral changes, for example, wearing masks and using air filters can reduce vulnerability to infection, while some other behaviors may increase their likelihood of getting infected. Thus, air pollution can affect disease transmission through these behavioral channels, as well.

Not surprisingly, both laboratory and epidemiological studies have shown that aerosol transmission can be an important mode of transmission of SARS (Yu et al., 2014), influenza (Clay et al., 2018; Graff Zivin et al., 2023), tuberculosis (Escombe et al., 2007), chickenpox (Leclair et al., 1980), and SARS-CoV-2 (i.e., COVID-19).

3. Empirical Strategy

3.1. The SIRD model

Our empirical analyses are based on the Susceptible-Infectious-Recovered-Deceased Model (SIRD Model), a model that is widely used by epidemiologists to depict the evolution of various infectious diseases. In this model, the individuals are classified into either Susceptible (S_t), Infected (I_t), Recovered (R_t), or Deceased (D_t). The size of each group evolves as follows:

$$S_t^{'} = -\beta_t S_t I_t \tag{1}$$

$$I_t^{'} = (\beta_t S_t - \gamma - \rho) I_t \tag{2}$$

$$R_t^{'} = \gamma I_t \tag{3}$$

$$D_t^{'} = \rho I_t \tag{4}$$

where β_t is the transmission rate, γ and ρ are the time-invariant recovered rate and the deceased rate, respectively. Our empirics aim to recover how air pollution affects the transmission rate (β_t), which is a widely used parameter to measure the spread of the epidemic because it deterministically affects disease development. Here, the reproduction number (R_{0t}) is proportional to the transmission rate ($R_{0t} = \beta_t/(\gamma + \rho)$).

In our analyses, we assume the proportion of the susceptible population is close to 1 ($S_t \to 1$) i.e., almost the entire population can be thought of as susceptible. This assumption is likely to be reasonable if infectious diseases are not prevalent or before proceeding to the serious pandemic. With this assumption, using equation (2), we can write the active epidemic growth as:

$$I_t^{'} = g_t I_t = (\beta_t S_t - \gamma - \rho) I_t = _{S_t \to 1} (\beta_t - \gamma - \rho) I_t \tag{5}$$

The solution of equation (5) can be described by the following exponential function:

$$\frac{I_t}{I_{t-1}} = e^{\beta_t - \gamma - \rho} = e^{g_t} \tag{6}$$

where I_t is active infections at time t, and the infection growth rate (g_t) is proportional to the transmission rate (β_t). Taking the natural logarithm of this equation and parameterizing the growth rate, we get

$$\log(I_t) - \log(I_{t-1}) = g_t = g_0 + \alpha^* f(AP_t) + \varepsilon_t \tag{7}$$

$$where, g_t = \beta_t - \gamma - \rho$$

We expect ambient air pollution to alter the growth rate (g_t) by influencing the virus transmission rate (β_t), either through altering

susceptibility or transmissibility, or both. Thus, we model the growth rate as a function of air pollution exposure $f(AP_t)$ and its average treatment effect (α), in addition to the baseline growth rate (g_0), which measures the growth rate without any exposure to air pollution, and a mean-zero error term (ε_{it}). If the recovery rate (γ) and the deceased rate (ρ) are time-invariant, using the growth rate of new confirmed cases should generate identical results.

The selection of outcome variables should be derived from the research question of interest. In our paper, we use the growth rate as the outcome variable because it provides insight into how pollution affects the transmission rate, which is our primary parameter of interest. In contrast, we do not use new confirmed cases ($C_t - C_{t-1}$) as an outcome because they are equal to $(\beta_t S_{t-1})I_{t-1}$, meaning they depend not only on the transmission rate (β_t), but also on the current level of disease prevalence (I_{t-1}).

3.2. The Flexible Distributed Lag model

To account for the dynamic relationship between air pollution and disease transmission, we estimate a set of Flexible Distributed Lag (FDL) models. This FDL model includes multiple lags of the independent variable in the regressions, allowing us to capture the delayed impacts due to the incubation, testing, and reporting period. For instance, the incubation period is suggested to be 2 days ranging from 1 to 4 days for influenza (WHO, 2023) and 6 days ranging from 2 to 13 days for COVID-19 (He et al., 2020b,; Li et al., 2020).

Estimating the dynamic impacts of air pollution is challenging because it exhibits high autocorrelation and significant volatility. As a result, simply estimating the coefficients of all lagged pollution can be imprecise, particularly when one uses high-frequency data. To address this challenge, the FDL model allows the lagged coefficients to vary across times by approximating the coefficients as a cubic B-spline function (Barwick et al., 2023).

Specifically, our econometric framework has the following specification:

$$g_{it} = \log(I_{it}) - \log(I_{i,t-1}) = g_{i0} + \sum\nolimits_k AQ_{i,t-k} * f(\alpha^z_{t-k}) + \varepsilon_{i,t} \tag{8}$$

where g_{it} is the observed daily growth rate of active infections in city i at time t and g_{i0} is the base growth rate in city i. $AQ_{i,t-k}$ denotes the air quality in city i at time t-k. α_{t-k} estimates the effect of air pollution in time t-k on the virus growth rate in time t. The FDL model approximates the set of coefficients α_{t-k} as a cubic B-spline function with z segments, as denoted by $f(\alpha^z_{t-k})$.

For illustration, suppose z = 1. The B-splines with one segment are simply equivalent to 3rd order polynomial:

$$f(\alpha^{z=1}_{t-k}) = \gamma_0 + \gamma_1 k + \gamma_2 k^2 + \gamma_3 k^3 \tag{9}$$

where the contemporaneous impact ($k = 0$) of air pollution $\alpha^{z=1}_t$ is γ_0, one period lagged ($k = 1$) impact of air quality is $\gamma_0 + \gamma_1 + \gamma_2 + \gamma_3$, and k-period lagged impact of air quality is $\gamma_0 + \gamma_1 k + \gamma_2 k^2 + \gamma_3 k^3$. Given this, we have

$$\begin{aligned} &\sum\nolimits_k AQ_{i,t-k} * f(\alpha^{z=1}_{t-k}) = AQ_{i,t} * \alpha^{z=1}_t + AQ_{i,t-1} * \alpha^{z=1}_{t-1} + \ldots + AQ_{i,t-k} * \alpha^{z=1}_{t-k} \\ &= AQ_{i,t} * (\gamma_0) + AQ_{i,t-1} * (\gamma_0 + \gamma_1 + \gamma_2 + \gamma_3) + \ldots + AQ_{i,t-k} * (\gamma_0 + \gamma_1 k + \gamma_2 k^2 + \gamma_3 k^3) \\ &= \gamma_0 * v_0 + \gamma_1 * v_1 + \gamma_2 * v_2 + \gamma_3 * v_3 \end{aligned} \tag{10}$$

where $v_0 = AQ_{i,t} + AQ_{i,t-1} + AQ_{i,t-2} + \ldots + AQ_{i,t-k}$, $v_1 = AQ_{i,t-1} + 2^1 * AQ_{i,t-2} + \ldots + k^1 * AQ_{i,t-k}$, $v_2 = AQ_{i,t-1} + 2^2 * AQ_{i,t-2} + \ldots + k^2 * AQ_{i,t-k}$, and $v_3 = AQ_{i,t-1} + 2^3 * AQ_{i,t-2} + \ldots + k^3 * AQ_{i,t-k}$. Because v_0 to v_3 all depend on the function of the observed air quality level, the regression only requires estimating the four parameters to obtain the k+1 coefficients of $f(\alpha^{z=1}_{t-k})$. Following this procedure, we can estimate all the lagged coefficients using the corresponding regressions. In a more general case, we can replace the baseline function by rth-order polynomials and include multiple segments, where each segment is assembled as a piecewise polynomial of order r-1 via enforcing C^{r-2} continuity.

3.3. The instrumental variable approach

To estimate the causal relationship between air pollution and disease transmission, our framework employs an instrumental variable approach. It is well documented in the economics literature that air pollution levels are endogenous (Arceo et al., 2016; Chay and Greenstone, 2003; Cheung et al., 2020; Deryugina et al., 2019; Deschênes et al., 2017; c; Heft-Neal et al., 2020; Knittel et al., 2016). In the case of infectious diseases, such concerns are exacerbated because their prevalence could change individual mobility patterns and reduce social interactions. Business operations can also stop, and the government can launch strict health interventions that undermine economic activities, all of which can affect air pollution levels and confound the relationship.

In particular, we estimate the following 2-stage model:

$$\widehat{AQ}_{i,t-k} = AQ_{i0} + \sum\nolimits_j IV_{i,t-j} * f\left(\delta^z_{t-j}\right) + e_{i,t} \tag{11}$$

$$g_t = g_{i0} + \sum\nolimits_k \widehat{AQ}_{i,t-k} * f(\widehat{\alpha}^z_{t-k}) + \varepsilon_{i,t} \tag{12}$$

where equation (11) is the first stage in which we estimate the predicted change in Air Quality ($\widehat{AQ}_{i,t-k}$) derived from variations in

instrumental variables ($IV_{i,t-j}$). AQ_{i0} is the baseline air quality level in each city i that is not affected by the instrument. In the second stage, we estimate how the growth rate of infections (g_t) is affected by the predicted change in air quality ($\widehat{AQ}_{i,t-k}$). For this instrument to be valid, the predicted change needs to be orthogonal to the error term ($\varepsilon_{i,t}$). Because we incorporate the FDL and IV model, we use variations in instrumental variables and air quality from time $t-k$ to time t, and approximate the set of coefficients for the first stage ($f\left(\delta^z_{t-j}\right)$) and second stage ($f\left(\widehat{\alpha}^z_{t-k}\right)$) as a cubic B-spline function with z segments.

4. Air pollution and first wave of COVID-19 outbreaks in China

4.1. Methods and data

We first apply the econometric model discussed above to evaluate the relationship between ambient air pollution and COVID-19 in China. We start with using OLS to estimate:

$$g_{it} = \log(I_{it}) - \log(I_{i,t-1}) = \sum\nolimits_k AQI_{i,t-k} * f\left(\alpha^z_{t-k}\right) + \sum\nolimits_k X_{i,t-k} * \sigma_k + \theta_i + \pi_t + \varepsilon_{i,t} \tag{13}$$

This equation is obtained by augmenting equation (8). Notations i and t are prefectural city and day, and g_{it} and $AQI_{i,t-k}$ are the COVID-19 growth rate and the Air Quality Index. Because the median incubation period for COVID-19 is about 6 days but ranges from 2 to 13 days (He et al., 2020b; Li et al., 2020), we include time-varying variables (AQI and control variables) during the period between $k = 0$, and $k = 21$ (previous 3 weeks) and estimate the effect $f\left(\alpha^z_{t-k}\right)$ starting from $k = 2$ to $k = 13$. In our primary model, we adopt z = 3 segments and z = 2 and z = 4 for the robustness check.

Our regression includes temperature, precipitation, wind speed, and snow depth as control variables ($X_{i,t-k}$) because these climatic conditions could affect virus transmission (Carleton et al., 2021). Additionally, θ_i and π_t denote city and date fixed effects, which are a set of city-specific and time-specific dummy variables. The city fixed effects (θ_i) account for time-invariant base growth rate (g_{i0}) in each city, which is affected by the city's income level, natural endowments, and short-term industrial and economic structure. The time fixed effects (π_t) account for shocks that are common to all cities on a given day, such as national virus containment policies, macroeconomic conditions, and the national air pollution time trend. Conditional on these fixed effects and the time-varying control variables, we assess how air pollution affects virus growth rate. If air pollution accelerates virus transmission, α_{t-k} will be positive.

To deal with the endogeneity concern, we use thermal inversion, which has been used as the instrumental variable for air quality in multiple previous studies (e.g., Arceo et al., 2016; He et al., 2019). When a thermal inversion occurs, a layer of warmer air overlays a layer of cooler air in the atmosphere. Because warmer air has a lower density, the air pollutants emitted from the ground surface are "trapped," which eventually leads to higher levels of local air pollution. As this phenomenon results from complex meteorological processes, it provides quasi-random variation in ambient air pollution levels.

One concern about the instrumental variable is that thermal inversion may directly affect virus transmission. To check if such a possibility exists, we look into the scientific literature on virus transmission and cannot find any scientific theory that links thermal inversion with virus transmission directly. Moreover, medical research shows that the airborne virus's survival time in the air is often very limited. For example, van Doremalen et al. (2020) found that the COVID-19 virus could only live for 3 h in the air. Given its short life span, the virus itself should not be able to travel long distances in the same way as the air pollutants. In other words, thermal inversion is unlikely to be able to substantially change the virus's mobility and ability to infect people, except through its impact on air pollution, which can affect individual susceptibility and/or the transmissibility of the virus.

We use a 2-stage least square (2SLS) procedure to estimate the IV model:

$$\widehat{AQI}_{i,t-k} = \sum\nolimits_j TInversion_{i,t-j} * f\left(\delta^z_{t-j}\right) + \sum\nolimits_j X_{i,t-j} * \chi_j + \tau_i + \varphi_t + e_{i,t} \tag{14}$$

$$g_{it} = \log(I_{it}) - \log(I_{i,t-1}) = \sum\nolimits_k \widehat{AQI}_{i,t-k} * f\left(\alpha^z_{t-k}\right) + \sum\nolimits_k X_{i,t-k} * \sigma_k + \theta_i + \pi_t + \varepsilon_{i,t} \tag{15}$$

where $TInversion_{i,t-j}$ in first-stage equation (14) denotes the difference in temperature between the surface ground and upper layer in city i in time t-j ($0 \le j \le 21$), with the larger number representing the warmer temperature in the upper layer (Therefore, we expect δ^z_{t-j} to be positive.). In the second stage in equation (15), we regress the daily virus growth rate on this predicted pollution concentration ($\widehat{AQI}_{i,t-k}$) rather than on observed pollution concentration ($AQI_{i,t-k}$). $f\left(\alpha^z_{t-k}\right)$ measures the plausibly causal impact of air quality on virus transmission. We cluster the standard errors at the city level.

The COVID-19 data are retrieved from the China National Health Commission (CNHC). The data comprise newly infected, recovered, and death cases from January 20 to April 1, 2020 in 330 prefectural cities in China (See Fig. 1 and Appendix Table 1 for summary statistics). During this period, there were 49,982 cases and 2553 deaths in Wuhan and 30,441 cases and 727 deaths in other cities. In our baseline analyses, we exclude Wuhan city due to concerns about the city's data quality (Li et al., 2020). We describe the data in detail in Appendix A.

The air quality data are obtained from 1605 air quality monitoring stations covering all of the prefectural cities in China. These data are collected from China's National Urban Air Quality Real-Time Publishing Platform. We use the Air Quality Index, which is a comprehensive measure of air pollution: the index is constructed using $PM_{2.5}$, PM_{10}, SO_2, CO, O_3, and NO_2 concentrations, with a lower AQI meaning better air quality. The thermal inversion data are obtained from the MERRA-2, where we use the difference in

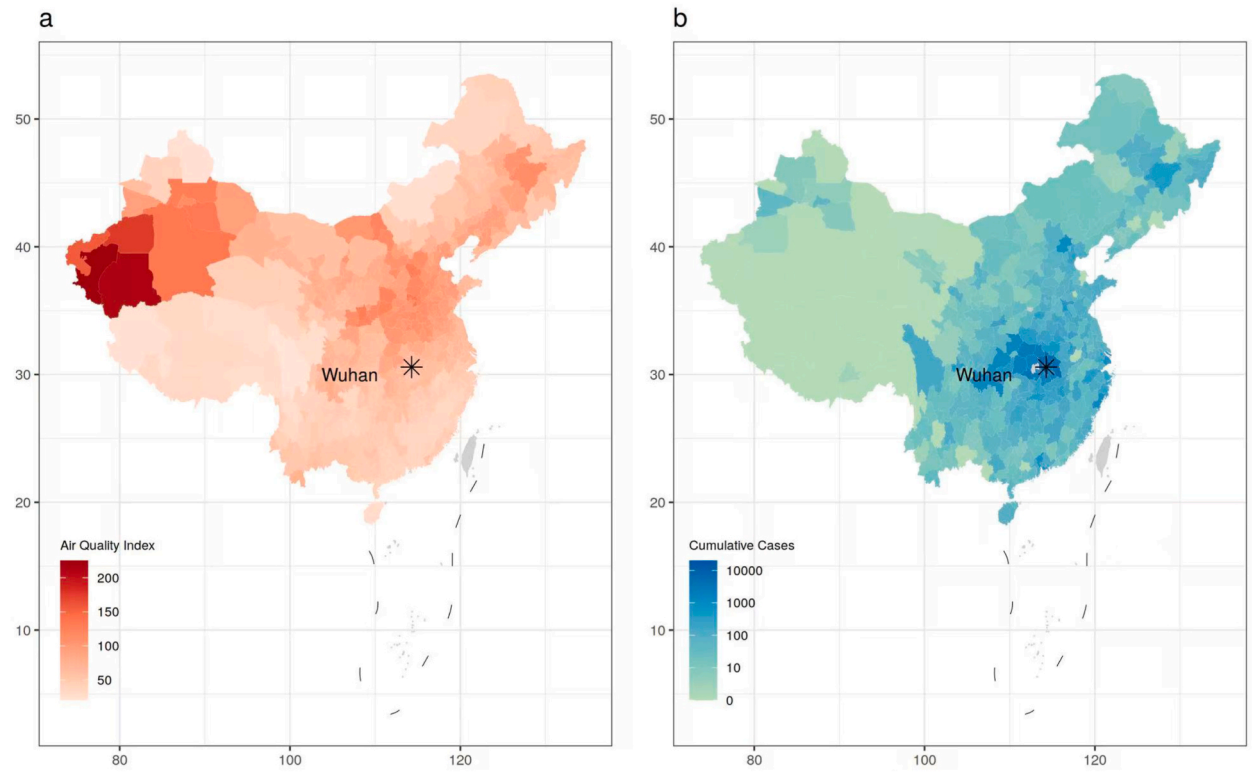


Fig. 1. Spatial Distribution of COVID-19 infections and air pollution in China.
Notes: Panel A shows the trend of the Air Quality Index (AQI). A higher AQI means worse air pollution. AQI is a comprehensive measure of air pollution: the index is constructed using $PM_{2.5}$, PM_{10}, SO_2, CO, O_3, and NO_2 concentrations (See Appendix Table 1). AQI is high in the Gobi Desert (Northwest part). Panel B shows the distribution of confirmed COVID-19 cases. The gray color denotes the no-data areas.

temperature between the first layer (110m) and the second layer (320m) following Arceo et al. (2016), Fu et al. (2021) and He et al. (2019). Lastly, weather data include temperature, precipitation, and snow depth, which were obtained from the Global Historical Climatology Network (GHCN) from the U.S. National Oceanic and Atmospheric Administration (NOAA).

In Fig. 2, we show that COVID-19 spread rapidly across the nation. The number of confirmed cases was close to zero on January 24 but reached 10,000 on February 2 and 20,000 on February 8, 2020 (outside Wuhan, Fig. 2A). From January 24 to February 11, the average growth rate of active cases was 13.77%, with a doubling time of 5.03 days (Fig. 2B). During our study period, the average AQI fluctuated and gradually declined over time, likely due to restricted economic activities.

4.2. *First-stage results: thermal inversion and air quality*

For the 2SLS procedure to be valid, the first assumption is that the instrument needs to be a strong predictor of the air pollution concentration. Fig. 3A shows that thermal inversion is indeed highly correlated with local air pollution levels, even after we include all the time-varying control variables and a set of fixed effects. Specifically, we find that a 1 °C increase in the temperature inversion is associated with a 2.83-point increase in the Air Quality Index (AQI), as shown in Fig. 3B. To confirm that this strong relationship is not driven by the local time trends or the spatial distribution of frequency of thermal inversions, we conduct two sets of permutation tests. Specifically, we randomly shuffle the observed thermal inversions (1) within the same location and (2) within the same day 1000 times. We find that the average estimates using the placebo sample in each case are close to zero, as shown in Fig. 3B. Lastly, the first stage F statistics is 36.08, which alleviates the weak instrument concern.[2]

[2] Our observed F-statistics are just above the traditional rule-of-thumb criteria (Andrews et al., 2024) but smaller than the more recent criteria proposed by Lee et al. (2022). This may be due to the use of the Flexible Distributed Lag (FDL) Model, in which we have multiple IVs (coefficients approximating the thermal inversion from t to t-k days) and multiple variables instrumented by these IVs (coefficients approximating the air pollution from t to t-k days). If we only use thermal inversion at time t to instrument AQI at time t, the reported F-statistics is 185. This suggests that our lower F-statistics are partially driven by practical reasons, rather than by a weak instrument problem.

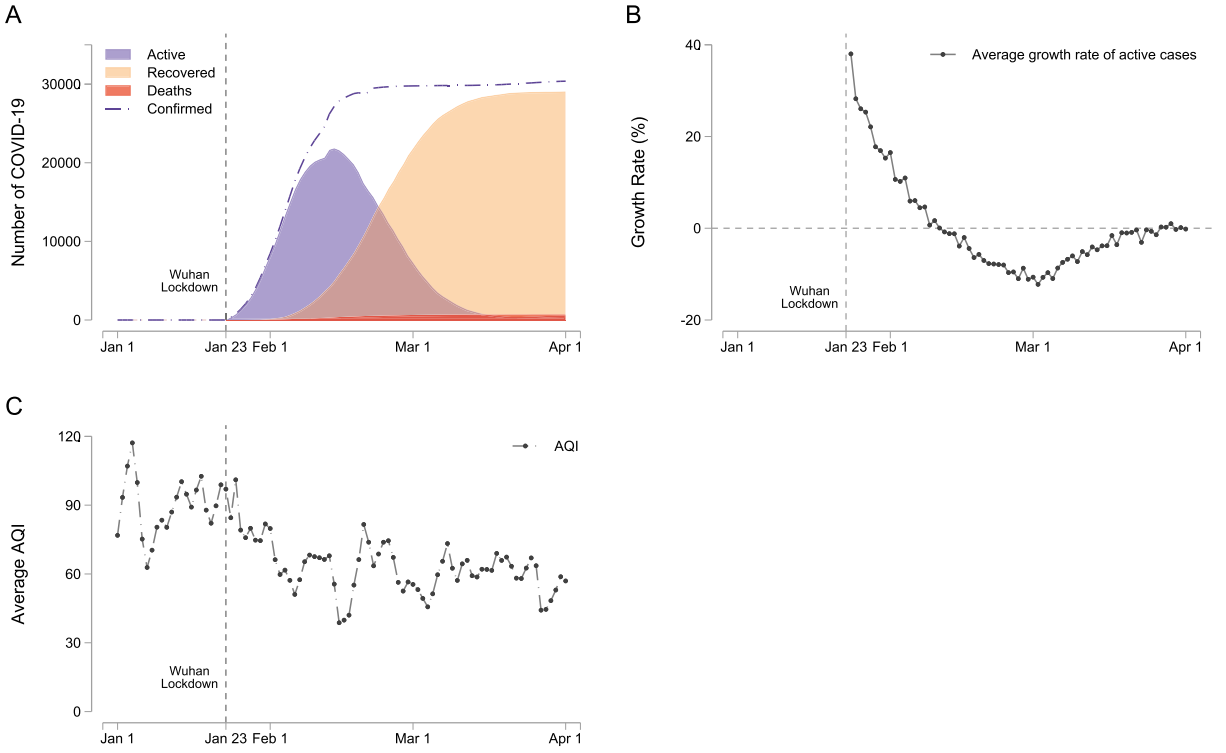


Fig. 2. Data on COVID-19 infections and air pollution in China.
Notes: In Panel A, we plot daily confirmed cases (blue dash line), active infections (blue area), recovered (orange area), and deaths (red area) from January 1 to April 1. The vertical line is the date of the Wuhan lockdown (23 January). Panel B plots the average growth rate of active infections. If it is larger than 0, the active infections increase. Panel C plots the trend of the Air Quality Index (AQI). A higher AQI means worse air pollution.

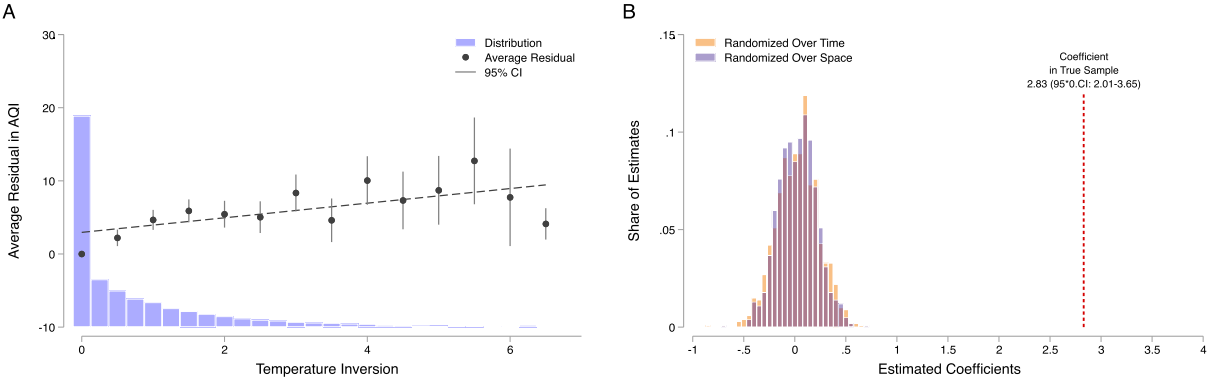


Fig. 3. Thermal inversion and Air Quality Index in China.
Notes: Panel A plots the distribution of thermal inversions, defined as the difference between the temperature in the second layer (320m) and that in the first layer (110m). The Y-axis represents the residual in variation in the AQI after controlling for temperature, precipitation, snow depth, city fixed effects, and date fixed effects. A higher temperature in the second layer is strongly associated with a larger residual in the AQI (worse air quality). In Panel B, we show that a 1 °C increase in temperature inversion is associated with a 2.83 increase in the AQI. We find no statistically significant association between placebo thermal inversions and the AQI. In the placebo sample, we randomized the thermal inversions across different days within the city or across different cities within a specific day.

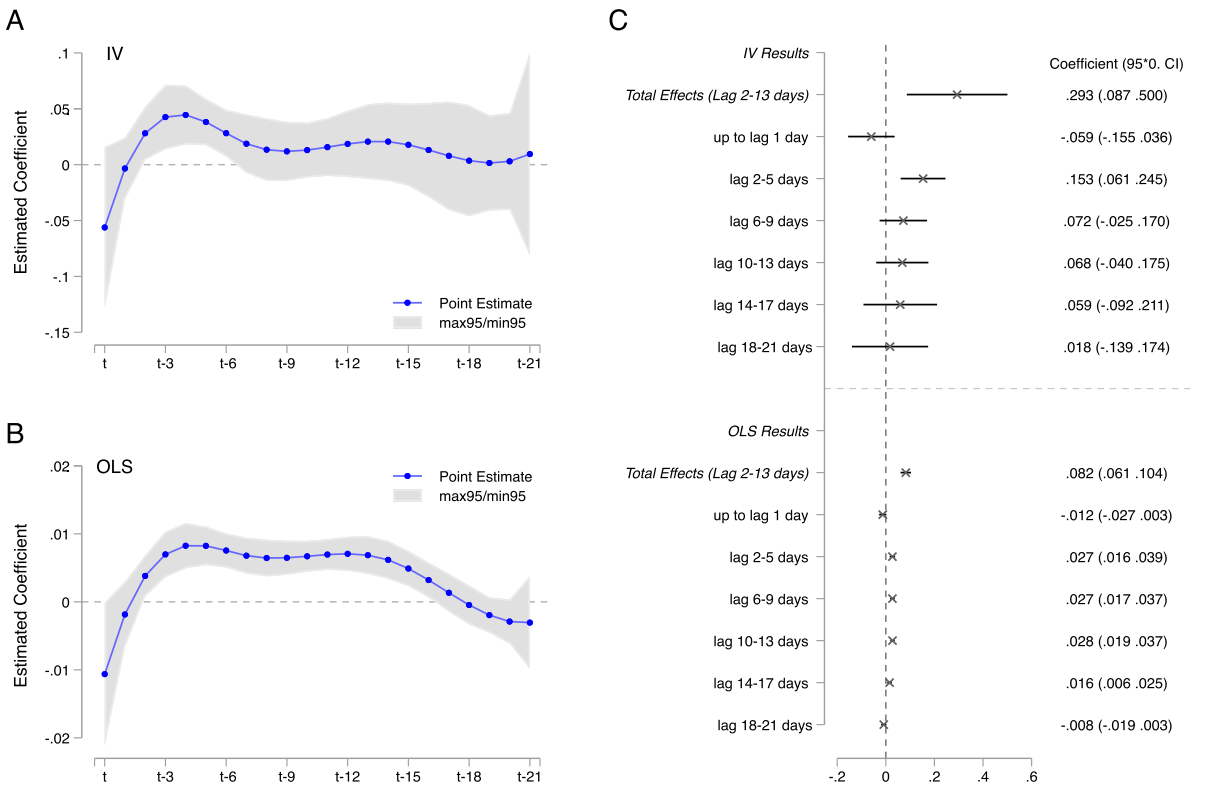


Fig. 4. Air pollution amplifies COVID-19 transmissivity in China.
Notes: Panel A plots the IV estimates. Weather controls (temperature, precipitation, and snow depth), date fixed effects, and city fixed effects are included in both the first and second stage regression. The blue dots represent the point estimates and the gray areas denote the 95% confidence intervals. Panel B reports the OLS estimates. The regression includes the same set of controls as in Panel A. Panel C on the right compares the magnitude of the effects between IV results and OLS results. The gray crosses represent the point estimates, and the gray lines show the 95% confidence intervals. In all regression, standard errors are clustered at the city level.

4.3. Second-stage results: dynamic relationship between air quality and COVID-19 growth rate

In the second stage, we use predicted air pollution from the first-stage regression to estimate the pollution-transmission relationship. While the regressions include up to 22 days of lags (current day + the previous 21 days), our preferred estimates focus on the joint coefficients for lags of 2–13 days. This choice is primarily guided by epidemiological evidence. We exclude estimates for exposure more than two weeks earlier ($t \geq 14$) because such exposures fall outside the typical incubation and reporting periods in China. Additionally, we exclude coefficients for $t = 0$ and $t = 1$, as it is nearly impossible to become infected, tested, and confirmed within two days of exposure. These edge periods also tend to be highly sensitive and difficult to interpret in the FDL model.

Our IV estimates are summarized in Fig. 4A (Table 2 for full results). We find statistically significant positive impacts of air pollution on COVID-19 growth rates with delays. Although the estimates for longer-lag exposures ($k \geq 7$) are noisier, the total effects for lags of 2–13 days are precisely estimated (Panel C, total effects). Specifically, a 10-point (14.3%) increase in AQI during this time window raises the growth rate by 2.93 percentage points. In contrast, outside the time window, air pollution does not have a meaningful effect.

These results are robust to a number of different model specifications. We add the city's lockdown status (Appendix Fig. 1A)[3] and days since the outbreak as control variables (1B), include Wuhan in our regression (1C), and use different FDL model segments (1D, 1E). All results remain similar. In addition, we add three days of future air pollution as another placebo test and find that, as expected, its effects are close to zero and statistically insignificant (1F). When we change the lengths of the lags of air pollution from 16 days to 24 days, we also observe similar patterns (Appendix Fig. 2).

We observe that the average daily growth rate in the first week of the epidemic outbreak is 24.9% across cities, with the doubling time of infections at 2.65 days. If AQI increases 10 points, the doubling time would be shortened to 2.38 days. Alternatively, assuming

[3] We use lockdown data from He et al., 2020a.

Table 2
The effect of air quality on the COVID-19 growth rate in China.

Dependent Variable: Daily disease growth rate	OLS		IV	
	coefficients	s.e.	coefficients	s.e.
	(1)	(2)	(3)	(4)
Current Day	−0.011**	[0.005]	−0.056	[0.037]
lag: 1 day	−0.002	[0.002]	−0.003	[0.014]
lag: 2 days	0.004**	[0.002]	0.028**	[0.012]
lag: 3 days	0.007***	[0.002]	0.043***	[0.015]
lag: 4 days	0.008***	[0.002]	0.044***	[0.013]
lag: 5 days	0.008***	[0.001]	0.038***	[0.011]
lag: 6 days	0.008***	[0.001]	0.028***	[0.011]
lag: 7 days	0.007***	[0.001]	0.019	[0.013]
lag: 8 days	0.006***	[0.001]	0.013	[0.014]
lag: 9 days	0.006***	[0.001]	0.012	[0.013]
lag: 10 days	0.007***	[0.001]	0.013	[0.012]
lag: 11 days	0.007***	[0.001]	0.016	[0.013]
lag: 12 days	0.007***	[0.001]	0.019	[0.015]
lag: 13 days	0.007***	[0.001]	0.021	[0.017]
lag: 14 days	0.006***	[0.001]	0.021	[0.018]
lag: 15 days	0.005***	[0.001]	0.018	[0.019]
lag: 16 days	0.003**	[0.001]	0.013	[0.021]
lag: 17 days	0.001	[0.001]	0.008	[0.025]
lag: 18 days	−0.000	[0.001]	0.004	[0.025]
lag: 19 days	−0.002	[0.001]	0.001	[0.022]
lag: 20 days	−0.003*	[0.002]	0.003	[0.022]
lag: 21 days	−0.003	[0.003]	0.009	[0.046]
z-Segment, k-order	3, 3		3, 3	
Observations (cities)	22,701 (329)		22,701 (329)	
Weather Control	Y		Y	
Date fixed effects	Y		Y	
City fixed effects	Y		Y	

Notes: The results correspond to Fig. 4. The dependent variable is the day-by-city level growth rate of the activated COVID-19 cases. Each estimate indicates the effect of the current and past air pollution (Air Quality Index) on the growth rate of COVID-19. Weather controls include temperature, precipitation, wind speed, and snow depth. Standard errors are clustered at the city level and shown in the right-side brackets. Significance levels are indicated by ***p < 0.01, **p < 0.05, and * p < 0.01.

the removal rate (total of recovered rate (γ) and deceased rate (ρ) in our SIRD model) is 17%∼20%, which means the patients recover or die in 5.8–7.5 days,[4] a 10-point increase in AQI would lead to 0.15–0.17 higher reproduction number ($R_0 = \beta/(\gamma + \rho)$).

Existing studies linking air pollution to various health outcomes often find that ordinary least squares (OLS) estimates understate the impact of air pollution (Appendix Fig. 3). We also find that the OLS estimates are substantially smaller than the IV estimates (even though the dynamic patterns are similar). OLS regression shows that a 10-point increase in AQI between 2 days and 13 days before the case is reported is associated with a 0.82 percentage point increase in the growth rate. This is about one-fourth of the IV estimate, suggesting that the OLS estimate can be biased downward. Appendix Figs. 2 and 4 provide more robustness checks using different model specifications.

4.4. Additional results: results by different air pollutants and death rate

We also provide the results using specific air pollutants ($PM_{2.5}$, PM_{10}, SO_2, NO_2, CO, O_3). Generally, we observe that higher pollution levels increase the growth rate of COVID-19 with 2–13 days of delays (Appendix Fig. 5A), specifically for $PM_{2.5}$ (A1), SO_2 (A3), and CO (A5). The only exception is ozone, for which we find a higher concentration of O_3 (A6) slightly decreases the disease transmission, even though the relationship is not statistically significant. This might be because (1) the concentration of ozone is often negatively correlated with other pollutants (Sillman, 1999), and (2) ambient ozone can inactivate the virus by disrupting the virus structure (Tseng and Li, 2006). Moreover, across most pollutants, OLS estimates (Appendix Fig. 5B) are substantially smaller than the IV estimates, consistent with our baseline findings.

We further investigate whether air pollution results in a higher death rate in Appendix B. Several studies suggest that air pollution increases deaths from COVID-19, likely due to its detrimental impact on the human's ability to recover from the infection. While this question has important policy relevance, our dataset does not provide us with sufficient statistical power to explore the relationship. This is because, outside of Wuhan, there were only around 700 deaths caused by COVID-19 in China, with 90% of the cities having no

[4] In our dataset, the individuals with infections are classified as recovered when they test negative, rather than when they recover from symptoms. Therefore, to estimate the reproduction number from our main results, we adopt the removal rate which is reported by existing studies. They suggest that an average interval is 5.8 days (He et al., 2020b) or 7.5 days (Li et al., 2020), making the removal rate 17.4%–19.6%.

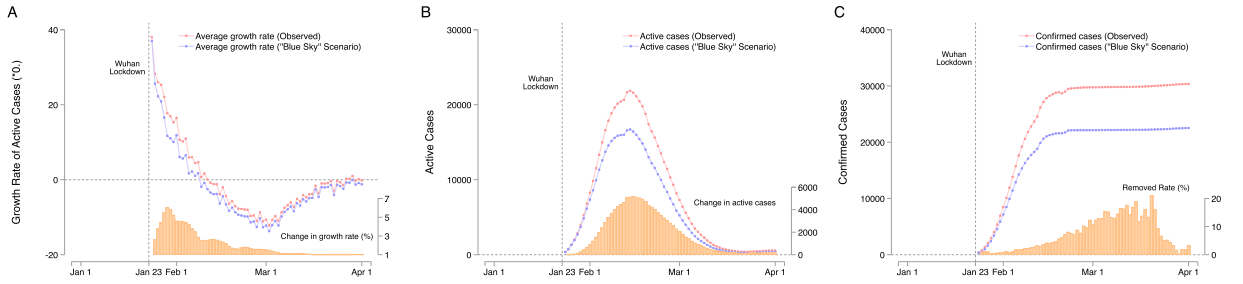


Fig. 5. Simulated impacts of air pollution on COVID-19 cases in China.
Notes: Panel A represents the observed daily growth rate of COVID-19 and the hypothetical rate in the "blue sky" scenario (an AQI exceeding 100 is replaced by 100). The orange bar represents the change in the growth rate (right axis). Panel B shows the active cases in the two scenarios and the orange bar shows the difference. Panel C predicts the estimated cumulative confirmed cases over time. To compute these numbers, we use the observed removal rate, which is shown by the orange bars. More technical details can be found in Appendix C.

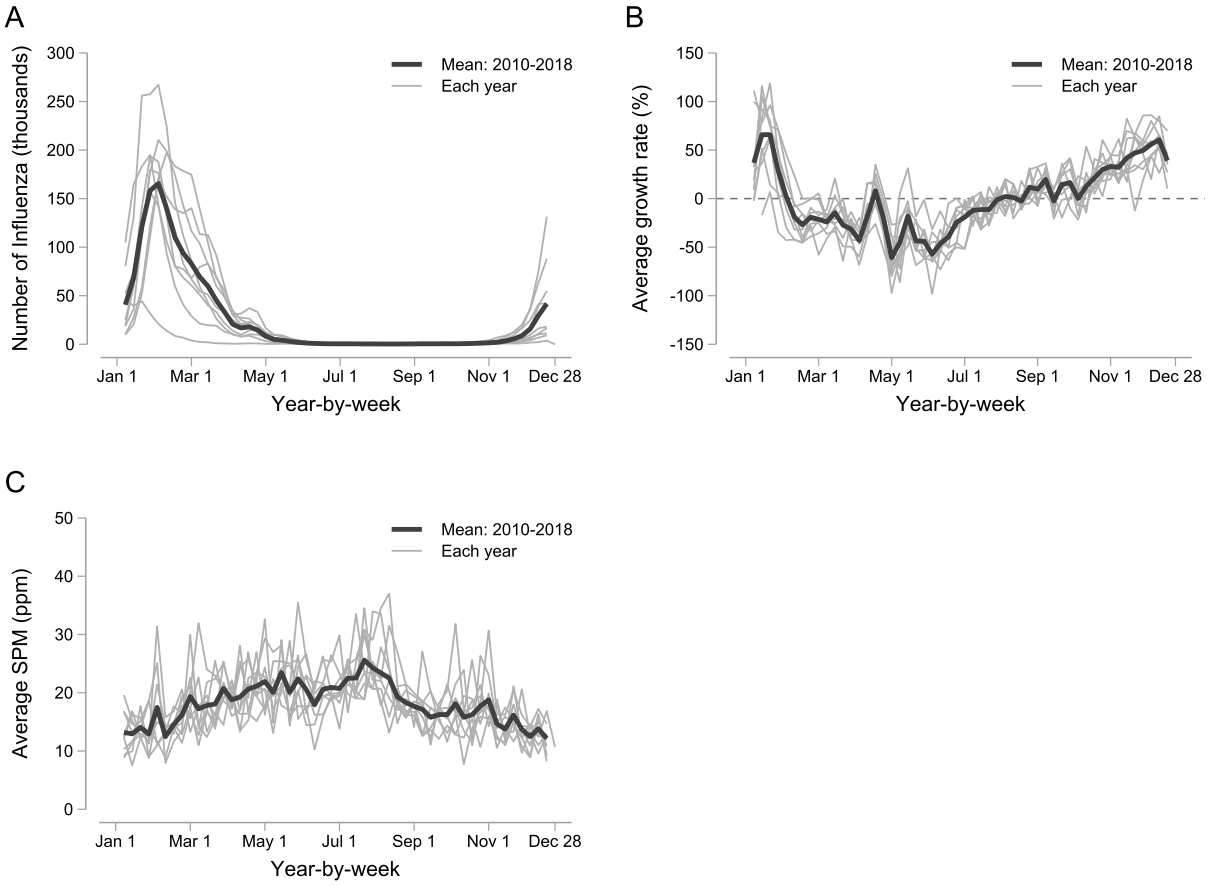


Fig. 6. Data on influenza and air pollution in Japan.
Notes: Panels A and B plot the weekly number of influenza detection cases and the growth rate, respectively. The bold line represents the mean, while the thin lines indicate the trend for each year. Panel C plots the year-by-week trend of suspended particulate matter (SPM) concentrations.

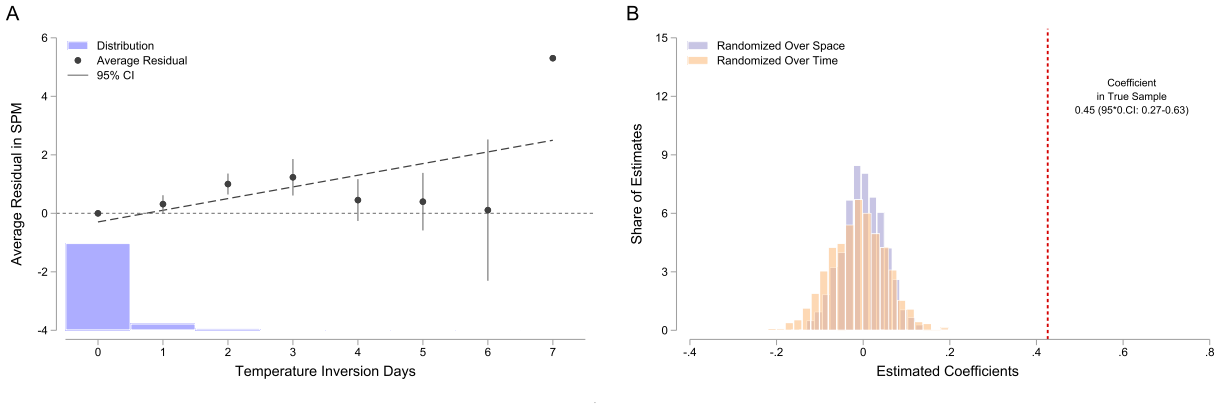


Fig. 7. Thermal inversion and Suspended Particulate Matter (SPM) in Japan.
Notes: Panel A plots the distribution of a weekly number of thermal inversion days, in which the thermal inversion day is defined as the day with a higher temperature in the second layer (320m) and that in the first layer (110m). The Y-axis represents the residual in variation in the SPM after controlling for temperature, precipitation, wind speed, prefecture fixed effects, and year-week fixed effects. A thermal inversion is strongly associated with a larger residual in the SPM (worse air quality). In Panel B, we show that one day increase in a weekly number of temperature inversion days is associated with a 0.47 increase in the SPM. We find no statistically significant association between placebo thermal inversions and the SPM. In the placebo sample, we randomized the thermal inversions across different weeks within the city, or across different cities within a specific week.

COVID-19 deaths or just one death.

4.5. *Back-of-the-envelope calculations for the "blue sky" scenarios*

We estimate the excess COVID-19 cases attributable to poor air quality (See Appendix C for details). In China, when daily AQI is below 100, it is considered as "good" or "moderate" air quality (also called a "blue sky" day). During our study period, 18.6% of the city-by-day observations do not meet this standard, and most of the "above-standard" (worse air quality) readings are obtained from northern Chinese cities. Here, we ask what would happen if we were able to bring the air quality index in all Chinese cities to below 100, holding other things constant.

Our estimates imply that the daily virus growth rate would have been slowed by 2.22% on average if all the cities had met the AQI = 100 standard (Fig. 5A).[5] Consequently, the number of active infections would have been reduced substantially. For instance, the number of active infections would have dropped from 21,855 to 16,457 (24.7%) on February 14, when we recorded the highest active infection number in China (Fig. 5B). Applying our estimates to the observed removal rate,[6] we expect that the cumulative confirmed cases would have been reduced by 30,376 to 22,181 during our study period (Fig. 5C).

The simulation shows that air pollution reshaped the exponential growth of the virus. We observe that the difference in the confirmed cases is small in the initial outbreak but gradually becomes larger as the outbreak progresses. If we do not model the virus transmission process using the SIRD model, we will not be able to capture the dynamic, cumulative, and non-linear impacts of air pollution on COVID-19 cases.

5. Air pollution and influenza in Japan

5.1. *Methods and data*

The econometric models for Japan's case are similar to those of China. To reduce redundancy, we describe the methods in detail for Japan's case in Appendix D. Here, we highlight three differences in econometric estimation when we study Japan's influenza. First, we use confirmed cases to construct our growth rate since the active cases are unavailable. Second, because the flu incubation period is typically shorter than that of COVID-19, ranging from 1 to 4 days, with symptoms developing 2–4 days after infection, we focus on the contemporaneous impact of air pollution. Finally, we use Suspended Particulate Matter (SPM), which measures microscopic solids or liquid droplets smaller than 10 μm since the AQI is not available in Japan.

In terms of data, we use prefecture-by-week data from 2010 to 2018 from approximately 5500 representative hospitals across 47 prefectures. We describe the data in detail in Appendix E (Appendix Table 2 for summary statistics).

[5] For the counterfactual analysis, we use results from $k = 2$ to $k = 13$, corresponding to the normal incubation period. We then estimate the counterfactual growth rate by applying this effect and using a linear approximation.

[6] For the back-of-the-envelope calculation, we use the observed removal rate to consistently project the epidemic growth in different policy scenarios.

Table 3
The effect of air pollution on influenza transmission in Japan.

Dependent Variable:	OLS		IV	
Weekly disease growth rate (%)	(1)	(2)	(3)	(4)
SPM (μg/m^3)	0.090	0.13	1.38	2.20
	(0.087)	(0.11)	(1.61)	(2.08)
SPM in t-1 week (μg/m^3)		−0.31*		−1.82
		(0.17)		(2.42)
SPM in t-2 week (μg/m^3)		0.23		−0.28
		(0.16)		(1.81)
Cumulative effects		0.055		0.096
		(0.12)		(1.65)
Observations	21,949	21,902	21,949	21,902
Control climate	Yes	Yes	Yes	Yes
Prefecture FE	Yes	Yes	Yes	Yes
Year-by-Week FE	Yes	Yes	Yes	Yes
First-stage F-statistics			77.6	20.9

Notes: This table reports the OLS and IV estimates of the pollution-transmission relationship for influenza. The dependent variable is the weekly growth rate of influenza, measured as the difference in the number of detection cases (in logarithm) between the current and last weeks. SPM denotes the concentration of suspended particulate matter. Significance levels are indicated by ***p < 0.01, **p < 0.05, and * p < 0.01.

Fig. 6 presents the trends in influenza cases and pollutant concentrations in Japan. During the sample period, there were, on average, 1.4 million cases annually. The influenza season typically starts from October to May and peaks in February (Panel A). We see a positive growth rate from October to January and a negative growth rate after that (Panel B). Panel C depicts the mean SPM concentration, where we see the seasonality, with its concentration highest in summer.

5.2. Results

The first-stage results show a strong association between thermal inversion and SPM concentration in Japan. Specifically, Panel A in Fig. 7 indicates that about 11% of our sample weeks experience at least one thermal inversion day. The occurrence of inversion is strongly associated with residualized SPM concentrations that account for variations in weather and fixed effects. In Panel B, we find that a one-day increase in the weekly number of thermal inversion days is associated with a 0.47 μg/m^3 increase in SPM concentration, significantly larger than estimates from our permutation tests. The associated F-statistics, ranging from 20 to 78, further substantiate the strong first-stage results.

The OLS results reveal weak and statistically insignificant associations between SPM concentration and the influenza growth rate, as shown in Table 3. We primarily focus on the contemporaneous relationship due to the shorter incubation and symptom period. A 1 μg/m^3 increase in SPM is associated with a 0.090 percentage point increase in the weekly growth rate of confirmed influenza cases (column (1)). This implies that a 10% increase in SPM (1.84 μg/m^3) is associated with a 0.17 percentage point increase in the weekly growth rate, suggesting that the magnitude is economically modest. The cumulative effects of up to two weeks even suggest a slightly weaker and statistically imprecise association (column (2)).[7]

The IV estimates reveal similar patterns with small and statistically insignificant results. While the IV coefficient is larger than the OLS coefficient, it is still imprecisely estimated. The IV results show that a 1 μg increase in SPM leads to 1.38 percentage point increase in the weekly growth rate (Column (3)), which corresponds to a 0.20 percentage point increase in the daily growth rate, a magnitude smaller than our COVID-19 coefficient. Adding more lags in the regression yields similar results (Column (4)). Further, including more flexible fixed effects (Column (1) in Appendix Table 3), adding the one-week lead of air pollution (Column (2)), and restricting samples to flu-prone seasons (Column (3)) all yield statistically insignificant results.

6. Conclusion

Estimating the relationship between ambient air pollution and airborne diseases is challenging because it requires researchers to account for virus epidemiological characteristics, to isolate the impact of air pollution from potential confounders, and to consider the potential delayed impact due to incubation, testing and reporting. We propose an econometric framework to address these challenges: we rely on the epidemiological Susceptible-Infectious-Recovered-Deceased (SIRD) model to construct the outcome of interest, the Instrument Variable (IV) model to estimate the causal relationship, and the Flexible-Distributed-Lag (FDL) model to understand the dynamics.

We apply this approach to two cases: COVID-19 data in China and influenza data in Japan. In the first case, we find that a 10-point (14.3%) increase in AQI leads to a 2.93 percentage point increase in the daily growth rate of COVID-19 infections over 2–13 days after

[7] We apply a distributed lag model, rather than "Flexible" distributed lag model. This adjustment is necessary because the polynomial approximation is challenging with only two weeks of lags.

the pollution exposure. The effect is statistically significant and economically meaningful. In comparison, when we apply our framework to assess how ambient air pollution influences influenza transmission in Japan, we fail to show that poorer air quality has a statistically significant impact on influenza growth rate.

The literature provided mixed findings on the relationship between air pollution and airborne diseases. In this study, we show that even if we use the same econometric framework, the results can still vary when using data from different countries. There are several reasons why the results between COVID-19 in China and influenza in Japan differ. First, the two diseases differ in various dimensions, including virus transmissibility, severity, vaccination prevalence, and herd immunity, which can lead to varying susceptibilities to air pollution. Second, the two cases are studied under different institutional settings: Japan and China have different levels of baseline environmental quality and socioeconomic characteristics of populations. For instance, the mean $PM_{2.5}$ concentration in Japan (13 $\mu g/m^3$) is about one-third of that in China (46.3 $\mu g/m^3$). Finally, the measurements, sample representativeness, and aggregation process of the disease data are also not identical across the two case studies, leading to slightly different econometric specifications. For example, Japanese data are based on subsets of representative hospitals and are aggregated weekly, which differs from Chinese COVID-19 data.

The findings, at least from China's case study, suggest that investments in air pollution control might yield health benefits beyond the well-documented impacts on physical health. When policymakers consider incorporating air pollution control into public health strategies, particularly during outbreaks of infectious diseases, it can potentially decrease the transmission rates of viruses, thereby alleviating the public health burden and economic costs associated with disease outbreaks.

We conclude by highlighting several limitations of this study and proposing directions for future research. First, while the SIRD model is useful for capturing the non-linear dynamics of virus spread, it relies on simplifying assumptions such as homogeneous population mixing and constant susceptibility, which may not fully reflect real-world complexities. Moreover, our assumption that air pollution primarily affects the transmission rate of airborne diseases, while having no impact on recovery or mortality rates, may not hold in reality. Future research should investigate how these assumptions influence the impact estimates. Second, our analysis focuses on two Asian countries, and the findings may not be generalizable to other regions with different environmental, social, and economic contexts. Future studies should expand the geographical scope to include more diverse settings, particularly in low- and middle-income countries where air pollution levels are often higher. Third, we use thermal inversion as an instrumental variable to address the endogeneity of air pollution. While this is a widely used approach in the literature, the validity of the instrument is ultimately untestable. Although there is no current scientific evidence indicating that thermal inversions affect virus survival and transmission through other channels, such a possibility theoretically exists. Future research should examine whether the results are robust to alternative instruments. Finally, due to data limitations, we are unable to explore the heterogeneity in the effects of air pollution on airborne diseases. It is important for future research to account for factors such as age, pre-existing health conditions, and socioeconomic status when estimating these effects.

Data and Code availability

Data and code used in this paper are available at https://github.com/yhecon/AirborneDiseases.

CRediT authorship contribution statement

Guojun He: Writing – review & editing, Supervision, Investigation, Funding acquisition. **Yuhang Pan:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Takanao Tanaka:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jeem.2025.103117.

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